

6. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

Fraction	Number of carbon atoms in fraction	Boiling point range (°C)	Colour of fraction	Flame when burning	Ease of burning
fuel oil	1-4	-170 to 20	colourless	clean	very easy
petrol	5-10	20 to 70	pale yellow	clean	easy
naphtha	8-12	70 to 120	yellow	some soot	quite easy
kerosene	10-16	120 to 240	dark yellow	smoky	quite difficult
diesel oil	15-30	240 to 350	brown	smoky	difficult

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

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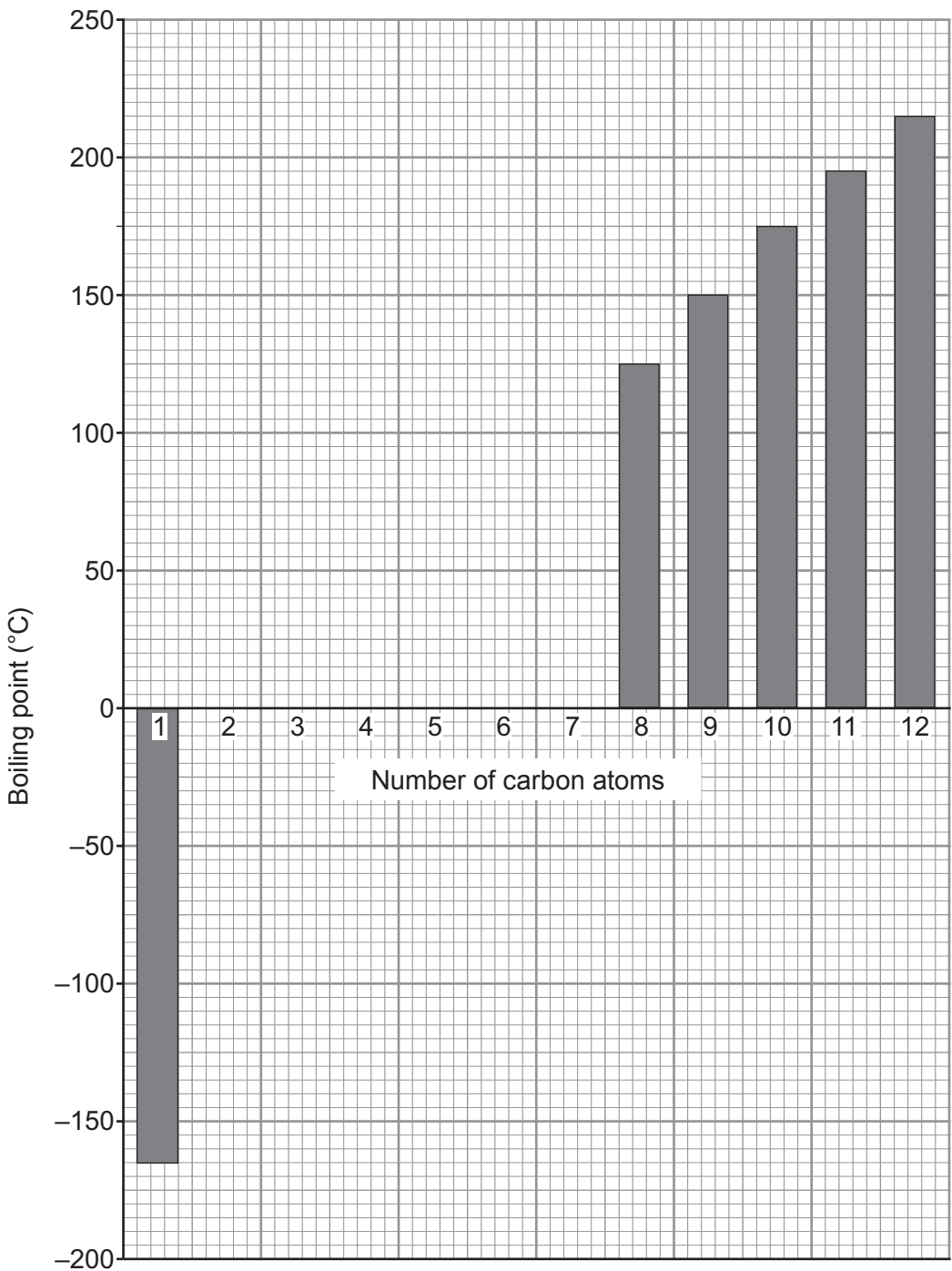
- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

Number of carbon atoms	Boiling point (°C)
1	-165
2	-90
3	-40
4	10
5	35
6	70
7	
8	125
9	150
10	175
11	195
12	215



Examiner
only

(i) Complete the bar chart below. Some of the bars have already been drawn. [2]



(ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method

Reason

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(ii) One of the hydrocarbons in petrol is pentane, C_5H_{12} .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

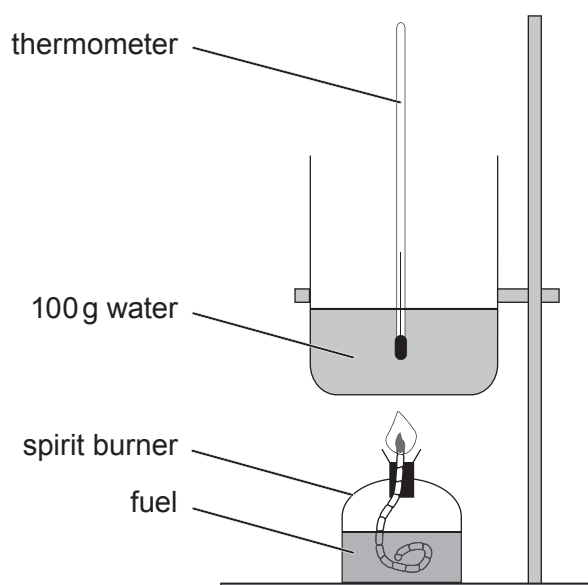
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\text{energy released per gram of fuel (J/g)} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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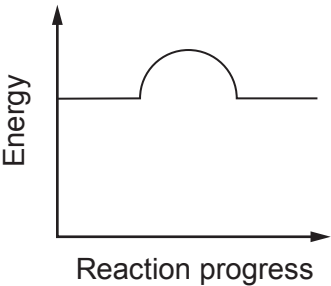
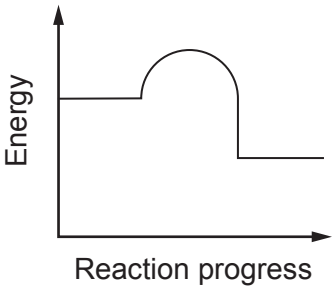
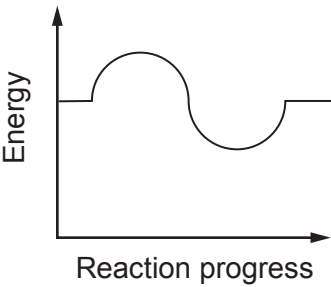
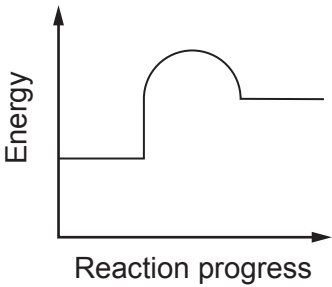
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

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(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



1 28

1	H	Hydrogen	1
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7	Li	Lithium	3	9	Be	Beryllium	4	11	B	Boron	5	12	C	Carbon	6	14	N	Nitrogen	7	16	O	Oxygen	8	19	F	Fluorine	9	20	Ne	Neon	10
23	Na	Sodium	11	24	Mg	Magnesium	12	27	Al	Aluminium	13	28	Si	Silicon	14	31	P	Phosphorus	15	32	S	Sulfur	16	35.5	Cl	Chlorine	17	40	Ar	Argon	18
39	K	Potassium	19	40	Ca	Calcium	20	45	Sc	Scandium	21	48	Ti	Titanium	22	51	V	Vanadium	23	52	Cr	Chromium	24	55	Mn	Manganese	25	56	Fe	Iron	26
86	Rb	Rubidium	37	88	Sr	Strontium	38	89	Y	Yttrium	39	91	Zr	Zirconium	40	93	Nb	Niobium	41	96	Mo	Molybdenum	42	99	Tc	Technetium	43	101	Ru	Ruthenium	44
133	Cs	Caesium	55	137	Ba	Barium	56	139	La	Lanthanum	57	179	Hf	Hafnium	72	181	Ta	Tantalum	73	184	W	Tungsten	74	186	Re	Rhenium	75	190	Os	Osmium	76
223	Fr	Francium	87	226	Ra	Radium	88	227	Ac	Actinium	89	204	Tl	Thallium	81	201	Hg	Mercury	80	197	Au	Gold	79	195	Pt	Platinum	78	210	Po	Polonium	84
												207	Pb	Lead	82	209	Bi	Bismuth	83	210	Po	Polonium	84	210	At	Astatine	85	222	Rn	Radon	86
												73	Ga	Gallium	31	75	As	Arsenic	33	79	Se	Selenium	34	80	Br	Bromine	35	84	Kr	Krypton	36
												112	Cd	Cadmium	48	106	Pd	Palladium	46	108	Ag	Silver	47	112	Cd	Cadmium	48	127	I	Iodine	53
												115	In	Indium	49	112	Cd	Cadmium	48	112	Cd	Cadmium	48	115	In	Indium	49	122	Sb	Antimony	51
												119	Sn	Tin	50	119	Sn	Tin	50	122	Sb	Antimony	51	127	I	Iodine	53	131	Xe	Xenon	54
												204	Tl	Thallium	81	201	Hg	Mercury	80	197	Au	Gold	79	195	Pt	Platinum	78	210	Po	Polonium	84
												207	Pb	Lead	82	209	Bi	Bismuth	83	210	Po	Polonium	84	210	At	Astatine	85	222	Rn	Radon	86

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number